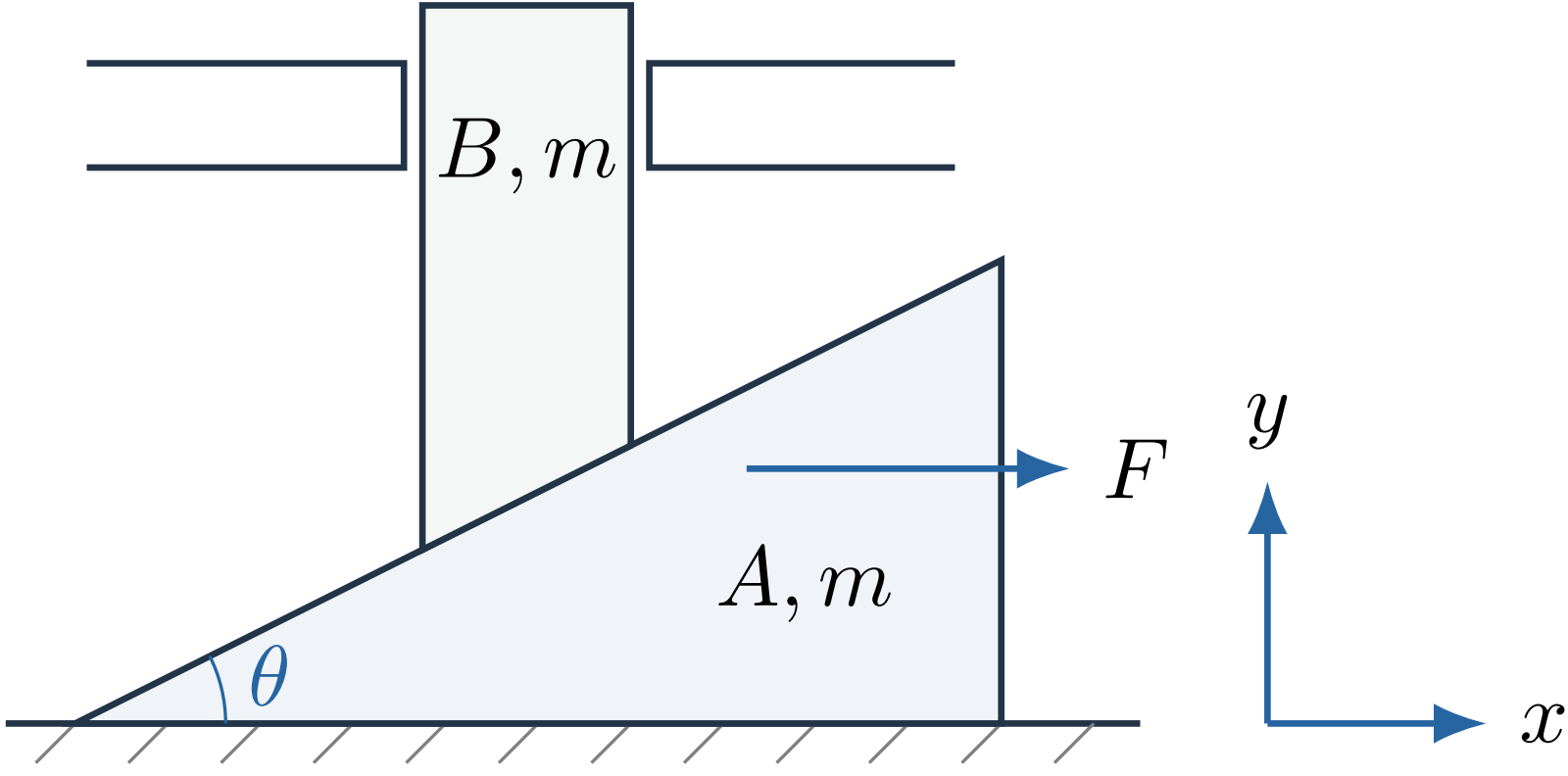光滑竖直导向